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Nordic Biobank × NVIDIA Federated Learning Hackathon for Biomedical Applications

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# Cogni Warriors

Predicting cognitive progression from routine phenotypes and imaging —  
**across institutions, without moving a single record.**

Multimodal late fusion · random-effects loss · NVIDIA FLARE 2.9 FedAvg · [open on GitHub](#)

## THE QUESTION

# Can low-burden measures predict executive function at every visit?

### The cohort

Penn LEAD (OpenNeuro ds007116) — 132 adolescents, 225 imaging sessions, up to 3 visits over ~1.5 years.

Three diagnostic groups. The modelling unit is the **session**, not the subject.

Inputs: 91 phenotype features plus an 80-dim structural-MRI embedding.

### The target

An executive-function composite: the mean of 5 z-scored Penn CNB tasks — N-back, PCET, AIM, CPT, Trail Making B.

Valid for **217 of 225 sessions** (at least 2 of 5 tasks passing QC).

Continuous and transdiagnostic — diagnosis is deliberately not the outcome.

Repeated visits → correlated observations

Data siloed across institutions

Centres instrument differently

No leakage: max |Pearson r| between any phenotype feature and the composite, or any of its 5 components, is 0.58.

## PREPROCESSING

# From raw scans and 42 phenotype files to one row per session

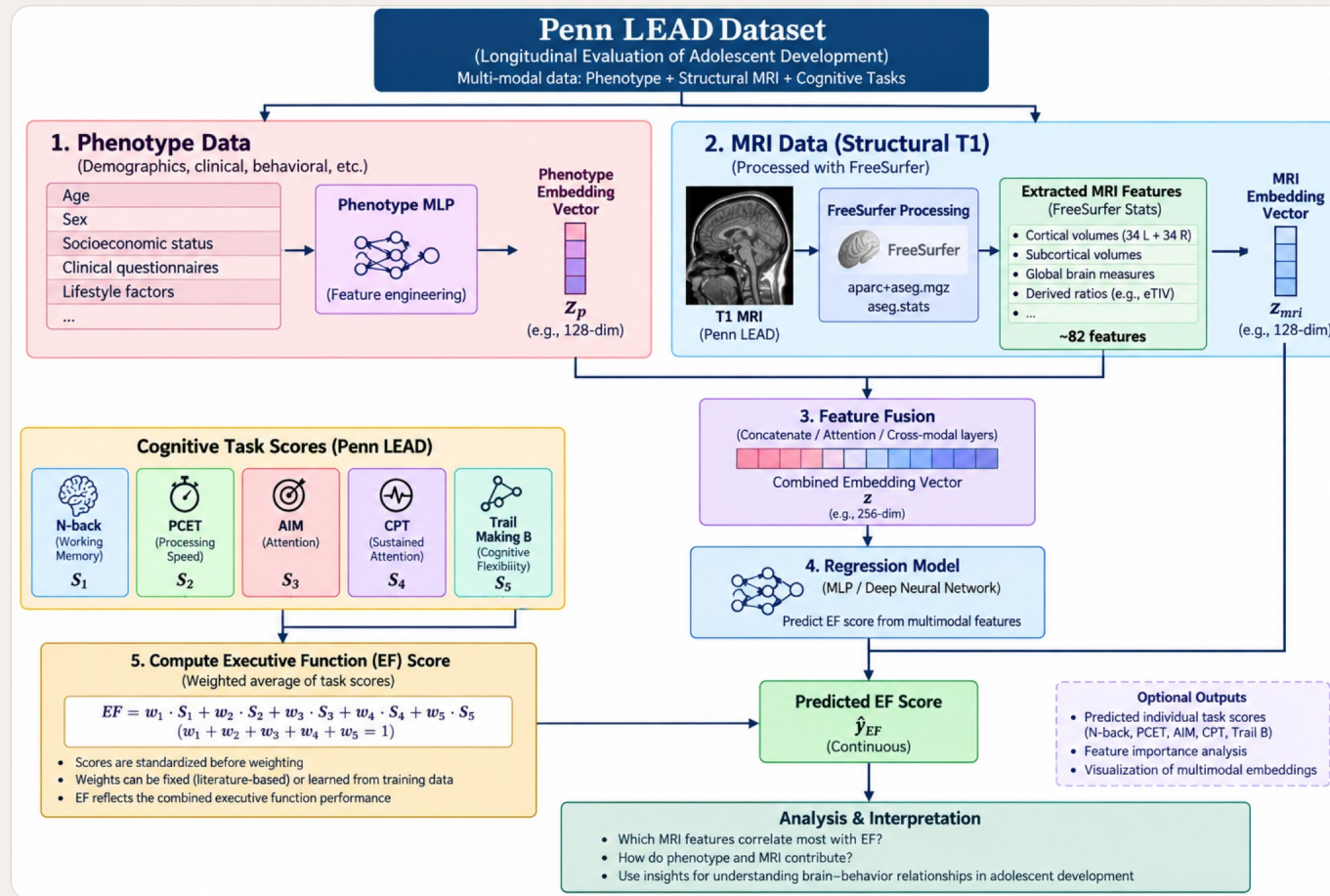
### Penn LEAD — Data Preprocessing Pipeline

From raw MRI scans to modeling-ready data with target



## THE PIPELINE

# Phenotype and imaging in, a continuous EF score out



### Two branches, one score

Phenotype and FreeSurfer-derived MRI features are embedded separately, then fused.

### EF is built, not borrowed

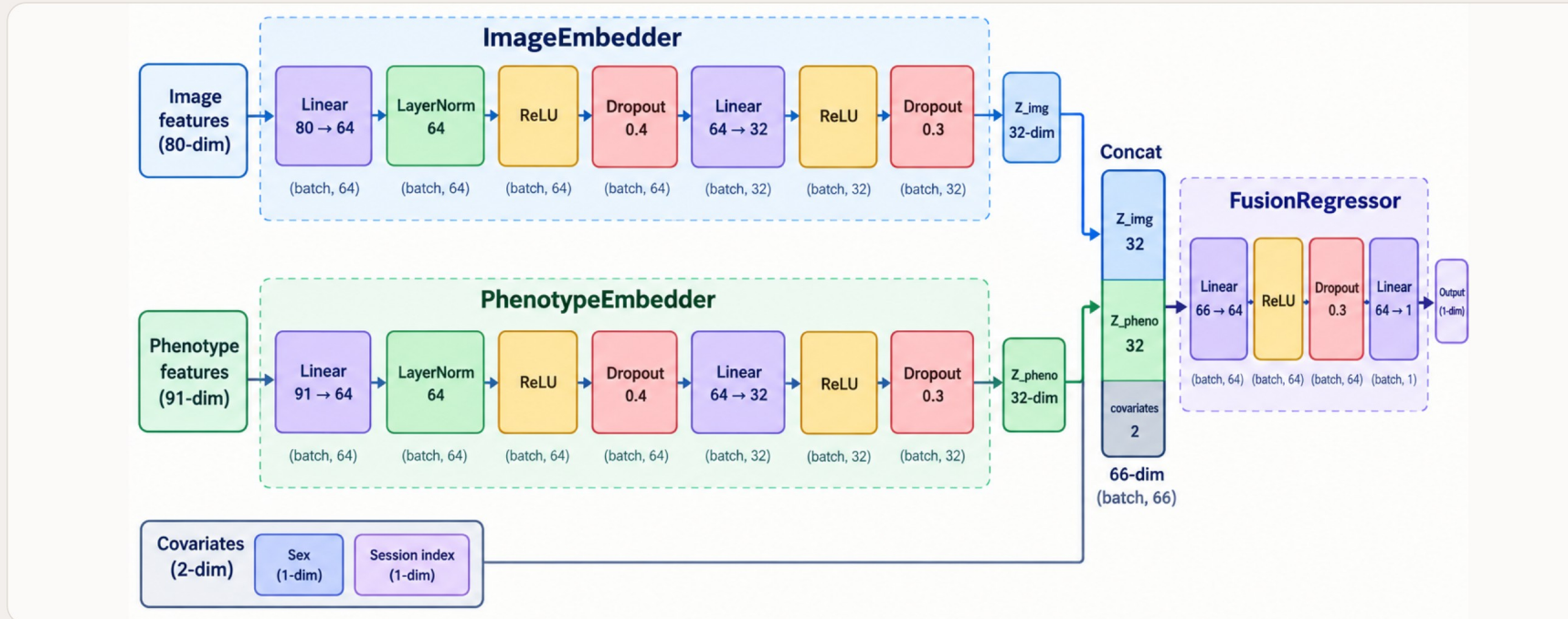
The five CNB task scores are standardised and averaged into the target — and held out of the inputs.

### Every block is swappable

Genotype or omics enters as one more branch; nothing downstream changes.

## THE MODEL

### Two encoders, one head, random effects in the loss



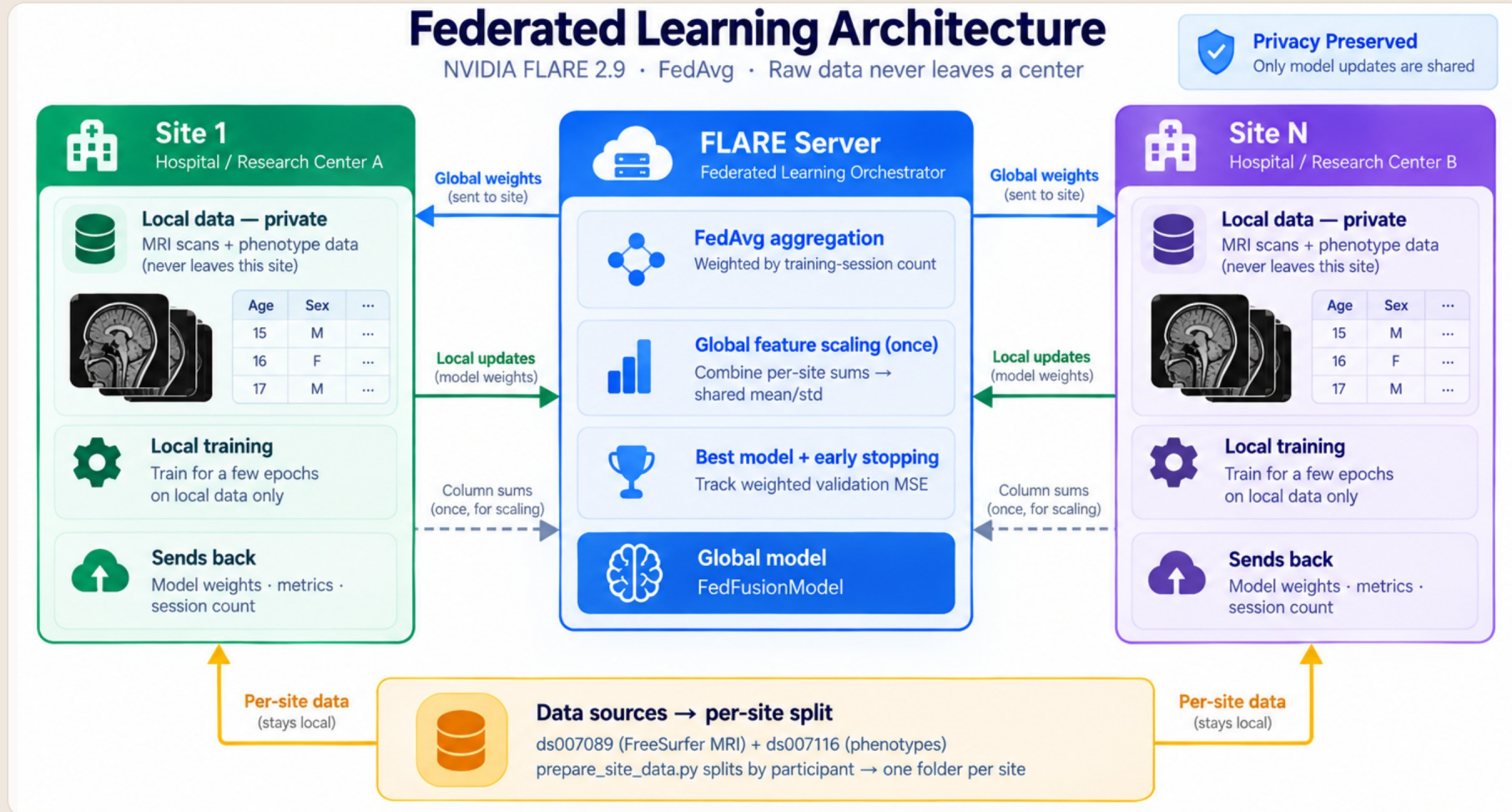
#### LMMNN loss — longitudinality, learned jointly

$$\text{loss} = \frac{1}{2} (r^T V^{-1} r + \log |V|), \quad V = \sigma^2_{\text{subject}} \cdot ZZ^T + \sigma^2_{\text{error}} \cdot I$$

#### Why it matters

A per-subject random intercept, optimised with the weights — no EM, no alternating steps. Subject-grouped minibatches; a subject's sessions never cross the split.

# Nothing but aggregates leaves a site



The server sees weights, metrics, session counts and one set of aggregate column sums — **never a record**. Runs as simulator, POC processes, an exported job, or a provisioned multi-machine deployment with Docker.

## RESULTS

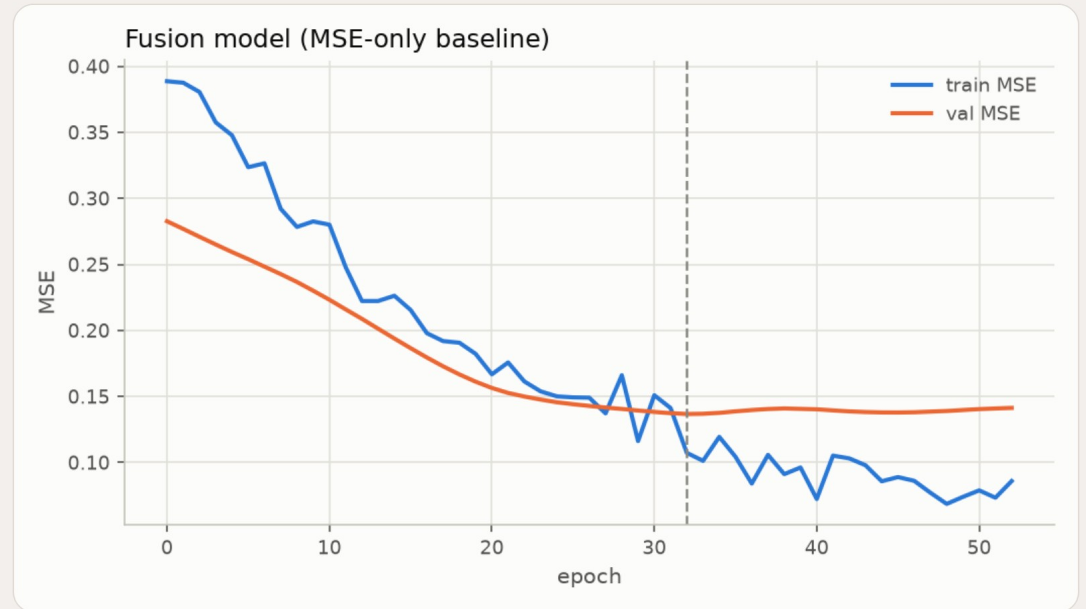
# FedAvg matches pooled-data performance

# 0.279 vs 0.275

Federated against pooled  $R^2$ , five seeds each — a gap far inside the seed-to-seed spread, and 2.2× what one site reaches alone.

| Setting                            | $R^2$             |
|------------------------------------|-------------------|
| Offline FedAvg, 5 seeds            | $0.279 \pm 0.039$ |
| Centralized pooled, 5 seeds        | $0.275 \pm 0.043$ |
| Single site alone, 5 seeds         | $0.126 \pm 0.038$ |
| NVFLARE simulator, 4 sites, seed 0 | 0.128             |

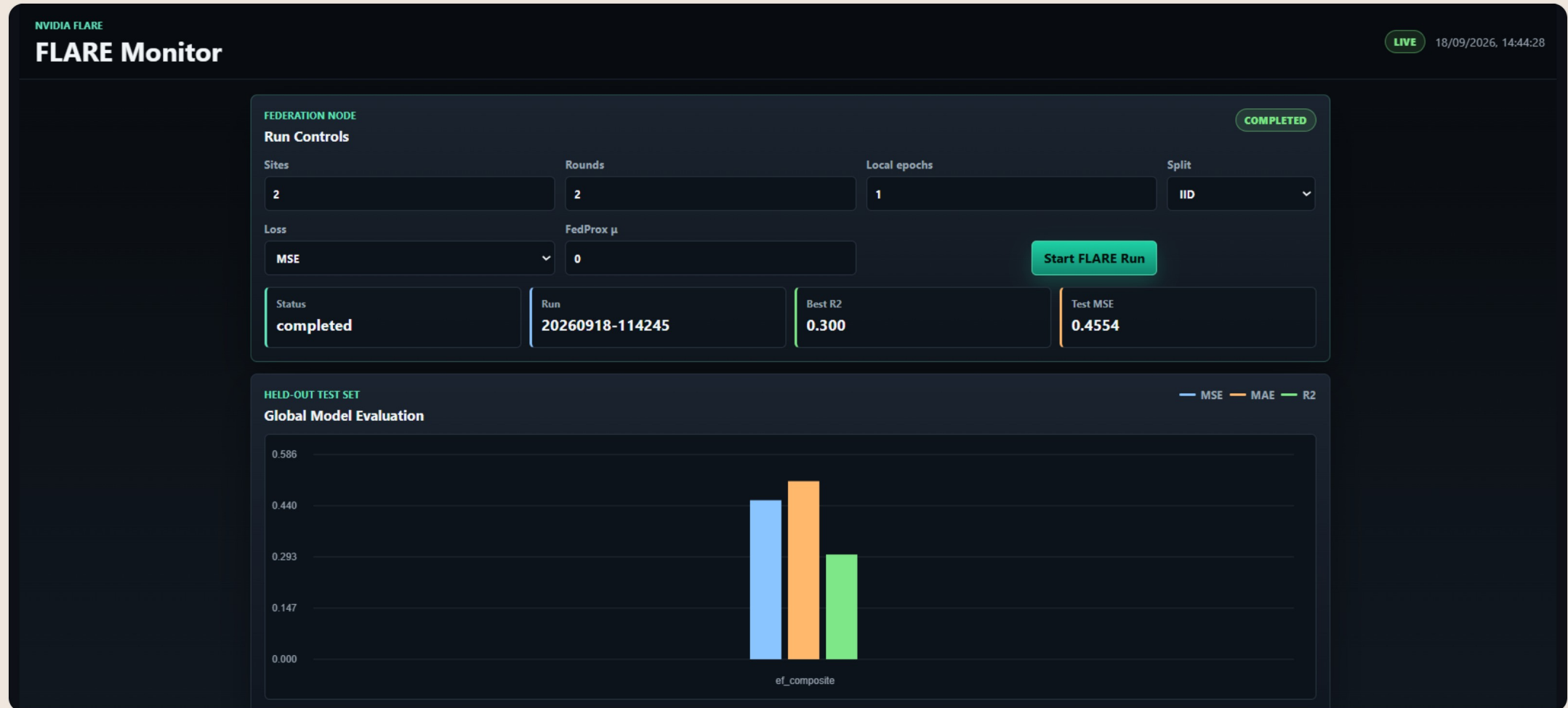
## Centralized fusion model — train vs. validation MSE



Target: EF composite, same 39-session held-out test throughout, held out by participant. IID 4-site split, MSE loss. LMMNN sanity run, seed 0:  $R^2 = 0.239$  against a mean-baseline MSE of 0.652.

IT RUNS

## One dashboard drives the whole federation



Sites, rounds, local epochs, IID or age-based split, MSE or LMMNN, FedProx strength — set them, press start, and the global model is evaluated on the held-out test set. Shown: a 2-site, 2-round run, best R<sup>2</sup> 0.300. Dockerised, one **compose up**.

## WHAT IT UNLOCKS

# A blueprint a Nordic biobank can adopt

### Modular

Swap the image embedder, the target (one composite or five sub-scores), or the loss — the rest of the pipeline adapts.

### Multimodal

Late fusion is a clean seam: genotype, omics or actigraphy enter as one more embedder into the same head.

### Cooperative

Global scaling from aggregates shows harmonisation is possible even when centres instrument differently.

### Longitudinal

Session-level rows with a per-subject random effect today; a site or scanner effect is the next natural term.

### Next

Real multi-machine deployment · per-site test sets · BLUP subject-specific correction at inference · a site/scanner random effect · a data-governance schema check. Code, docs and results are open on GitHub.